LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future

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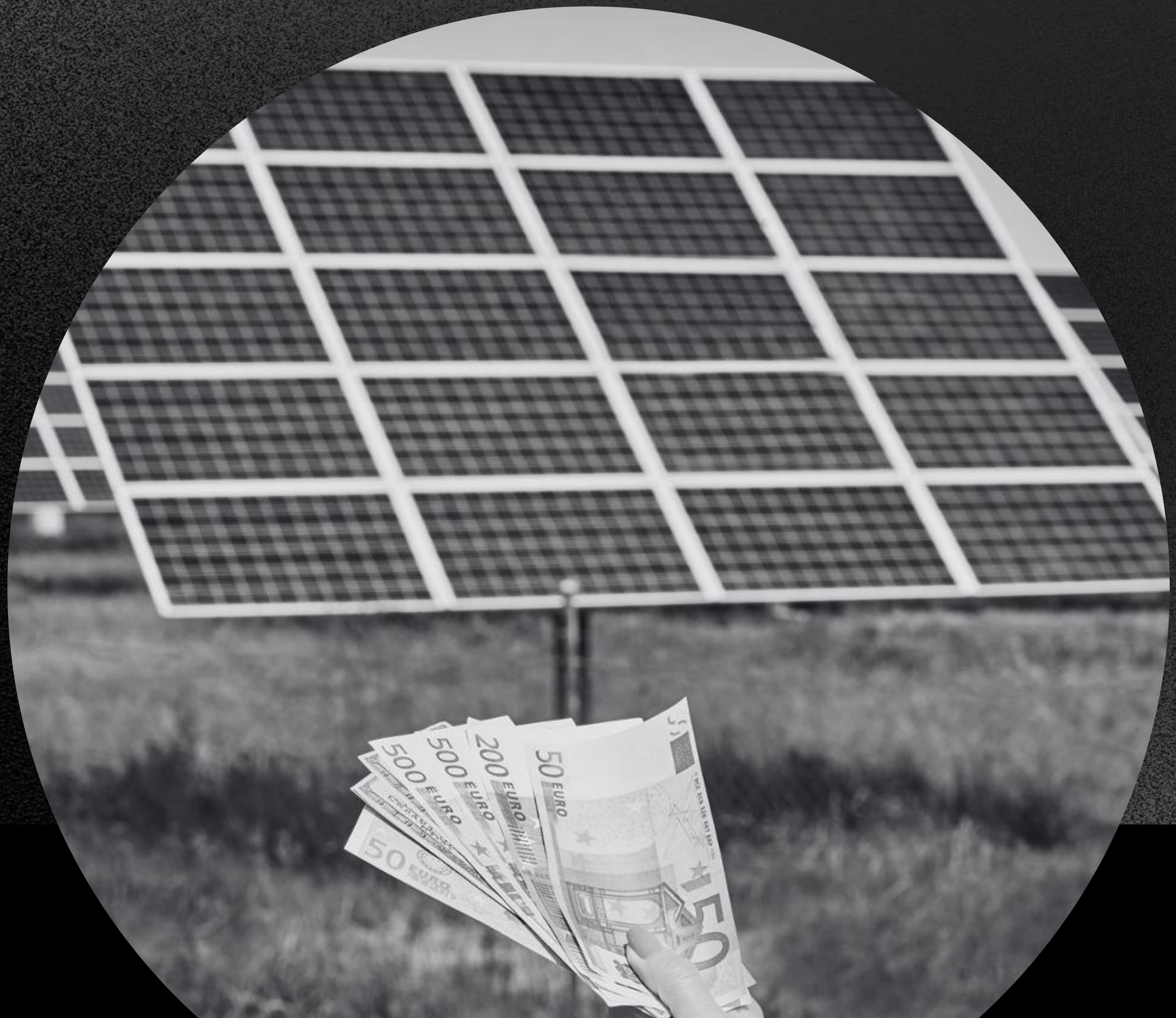
London Petro Academy Ltd

Solar Power Economics, Business Models & Finance

Designing and financing bankable solar PV projects across utility-scale, commercial and off-grid markets.



Course Overview

This intensive Solar Power Economics, Business Models & Finance course gives participants a practical, end-to-end understanding of how solar PV projects create value and secure funding across different market segments. It starts with solar technology fundamentals, including modules and cells, balance-of-system components, resource assessment and energy yield modelling, so participants can link technical choices to production and cost outcomes. The course then explores utility-scale, commercial/industrial and off-grid business models, covering tenders, merchant solar, self-consumption, net-metering, leases, community power and PAYG solutions for energy access. Building on this, delegates are introduced to project finance structures for solar, learning how to interpret cash-flow waterfalls, calculate DSCR and other coverage ratios, and understand how reserve accounts, covenants and credit-enhancement instruments impact debt capacity and equity returns through case-based group work.

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COURSE OBJECTIVES

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By the end of this course, participants will:

- Understand key technical and economic drivers of solar PV projects, including module and inverter choices, system design, resource characteristics and energy yield.
- Be able to compare and evaluate major solar business models (utility-scale, C&I and off-grid), and understand how technology and financial innovation are enabling new approaches.
- Learn core solar project finance concepts, including lender ratios, cash-flow waterfalls, DSCR/LLCR/PLCR, reserve accounts and their effect on debt capacity and equity returns.
- Recognise and manage key risks in solar project finance, including technology performance, resource uncertainty, policy changes, curtailment and payment/FX risks, using contracts and insurance.
- Gain hands-on experience reviewing solar financial models and testing sensitivities around energy yield, CAPEX/OPEX and PPA structures to optimise project bankability and returns.

Course audience

This course is designed for professionals who develop, finance, invest in or procure solar projects and need a solid, commercially grounded understanding of solar technology, business models and finance. It is particularly relevant for project developers and sponsors, OEMs, EPC contractors and O&M service providers, IPPs, utilities and grid operators active in solar markets. Private equity and institutional investors, development finance institutions and commercial lenders will benefit from the focus on risk, bankability and financial structuring. It is also suitable for corporate procurement and contract managers, government officials and regulators, and business developers and sales managers seeking to understand how different solar models can be structured and financed to meet commercial and policy objectives.

1

Solar PV Economics

PV Modules & Cells

- Silicon, ingots, wafer, cells to modules & solar PV power plants
- Silicon, thin film, multijunction, organic and concentrated photovoltaics
- Application – from handheld, to rooftop to gigawatt scale
- Watt-peak, cell efficiency, performance ratio and STC – Watt means what?
- Environmental temperatures and degradation
- Current cost trends and future innovations

Balance of System

- Balance of system including inverters, transformers, cabling and structural components
- One-axis and two axis tracking system – increased energy output or headache?
- System plant design and losses – from DC to AC
- Single string inverters versus central inverters
- The efficiency curve in function of rated output power
- Driving cost down for BoS – where are CAPEX and OPEX heading to

Resources & Energy Yield

- From global irradiation to direct, diffuse and horizontal irradiation
- Why the sun does not shine the same for polysilicon and thin film technologies ?
- AirMass, Clearness and Clearsky Index
- Using satellite and terrestrial data available for solar resource assessment
- Uncertainties of power curves or it wasn't like this in the brochure
- Understanding the resource assessment report - standard deviation, variance and p50/p90
- Experience of predictability and variability of solar resources assessments over time

PV System Modelling

- Group work to model a generic PV plant to examine the sensitivities of
- different resource data set from various locations in the world
- Module type and inverter choice
- Array size, degradation and shading in relation to overall annual energy production, performance ratio and losses

2

Solar PV Business Models

Utility Scale

- Falling CAPEX, short construction time, stable OPEX
- Solar tenders and around the world lessons learned
- Property cost and power density beyond of 100MW/km2
- Solar hedge against rising electricity tariffs
- Peak power and trading
- Emerging merchant power – the case for mining companies

CASE WORK: business & financial modelling of a 100MW solar PV plant

Grid Integration & Storage

- Micro grids, mini grids & smart grid – which to connect where?
- National energy transitions in Germany, China and US
- Growing curtailment threats in saturated energy markets
- energy storage as a potential game changer
- Electricity storage and distribution strategies

CASE STUDY: Economic impact of storage for a solar PV plant

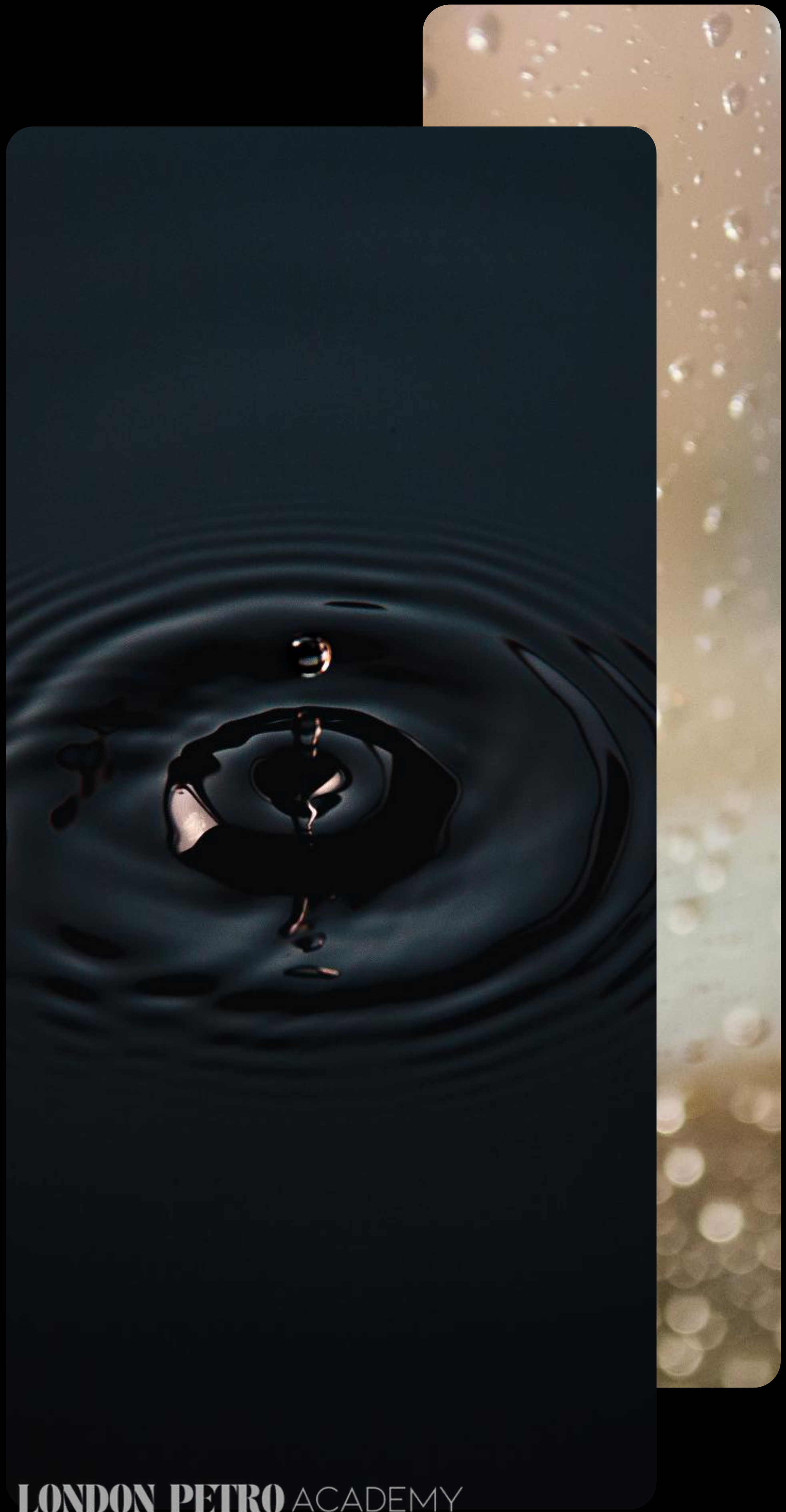
Private & Commercial

- The bottom-up revolution is happening now – market snapshots and learnings
- Self consumption and net metering
- Legal challenges for roof-tops and what about scale?
- Leases, Energy cooperatives, community power and crowdfunding – a new force
- Standardisations driving capital costs downs
- Efficient installations to reduce CAPEX

CASE WORK: business & financial modelling of a 100kW solar PV plant

OFF-GRID

- 20% of world population without electricity access – untapped market potential
- Plug& play solar home systems, lighting and phone charging
- How mobile payment drive off-grid applications
- Pay-As-You-Go Business Model to make PV Affordable
- The challenge of consumer finance and payment collection
- Solar & Diesel hybrids or substitution
- CASE STUDIES: M-KOPA and competitors – the quest for scalability



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3

Solar Finance

Debt Finance

- Project and bond finance within the renewable sector
- What means a bankable project?
- Understand lender's ratios
- Project financing structures
- Complex commercial and legal relationships
- Understanding the Cash Flow Waterfall
- Calculation of CFADS, DSCR, LLCR, PLCR
- How to drive overall debt capacity
- Reserve Accounts, Cash Sweeps, Cash Traps & Dividend Blockers
- Impact on equity returns

Risk Mitigation

- Multilateral banks, national development banks and export credit agencies
- Turn-key contract for engineering procurement and construction
- Equipment supply agreements – the question of liabilities and responsibilities
- O&M service contract
- availability, spare parts and reaction time
- Manufacturing warranties – typical coverage and what happens afterwards
- Off-take agreements for power, heat and certificates –the floor is the limit
- Risk instruments for construction delays, policy changes, curtailment and sun volatility

CASE WORK: Looking at sample contracts and polices to highlight key issues in risk management

Financial Analysis

Group work: Analyse financial models on different solar projects and understand the sensitivities of changing input parameters such as

- Energy yield, uncertainties and the bankability impact
- Capex & Opex
- PPAs level and duration

Financial Structuring

Group work to discuss solutions on a financial structure to optimize:

- DSCR, debt capacity and Debt:Equity ratio
- Equity returns and Net present value
- Cash-flow profile and payback

This course is available In-House and can be tailored to your specific requirements.

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Touchpoints

Question about this course?

Call us on +44 (0) 1582 516247 and one of our representatives will assist you with your enquiry.

3 Easy ways to register

Online: www.londonpetroacademy.co.uk

Email: info@londonpetroacademy.co.uk

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